Insurance Claims Fraud: Data Discoveries

What Senzing found in your data

Evidence comes from the SDK-generated query reports and the source-overlap assessment. Entity IDs and source-record keys are retained so each observation can be checked; candidate relationships are review leads, not fraud findings.

Headline numbers, interpreted

The three collected files contain 159 records: 120 POLICYHOLDERS, 22 VENDOR_REGISTRY, and 17 SIU_WATCHLIST. The mapped load contains 162 engine records because the VENDOR_REGISTRY mapping emits three embedded employer records. The SDK-backed export reconciled those 162 records into 90 entities. The How audit found 72 resolution steps across all 52 multi-record entities; these steps show how records joined engine entities, not 72 adjudicated identity truths. Fifteen entities span multiple sources, and the export reports 70 unique relationships. Eleven entities contain at least three records.

These are resolution outcomes and review signals, not measured accuracy. No adjudicated truth labels are available, so precision, recall, and F1 are not measured.

Merges and match keys

The table lists every resolution step returned by the SDK-backed How audit. Each side is the set of records in a virtual entity immediately before that step; repeated records are expected when a group grows through successive merges.

Entity	Step	Virtual entity 1 records	Virtual entity 2 records	Match key
20001	1	VENDOR_REGISTRY/2012	POLICYHOLDERS/2011	+NAME+ADDRESS
20002	1	POLICYHOLDERS/1069	POLICYHOLDERS/1070	+NAME+DOB+ADDRESS+NATIONAL_ID
20002	2	VENDOR_REGISTRY/2013	POLICYHOLDERS/1069; POLICYHOLDERS/1070	+NAME+DOB
20003	1	POLICYHOLDERS/1071	POLICYHOLDERS/1072	+NAME+DOB+ADDRESS+NATIONAL_ID
20003	2	VENDOR_REGISTRY/2014	POLICYHOLDERS/1071; POLICYHOLDERS/1072	+NAME+DOB
20004	1	VENDOR_REGISTRY/2041	POLICYHOLDERS/2042	+NAME+ADDRESS
20005	1	VENDOR_REGISTRY/2051	SIU_WATCHLIST/2052	+NAME+ADDRESS+PHONE
20006	1	VENDOR_REGISTRY/2061	SIU_WATCHLIST/2062	+NAME+ADDRESS+PHONE
20006	2	VENDOR_REGISTRY/2061; SIU_WATCHLIST/2062	POLICYHOLDERS/2063	+NAME+PHONE
20007	1	VENDOR_REGISTRY/2071	POLICYHOLDERS/2072	+LIKELY_NAME+ADDRESS+PHONE
20008	1	VENDOR_REGISTRY/2074	POLICYHOLDERS/2073	+NAME+ADDRESS
30001	1	VENDOR_REGISTRY/2101	VENDOR_REGISTRY/2102	+PNAME+PHONE+EMAIL
30003	1	VENDOR_REGISTRY/2111	VENDOR_REGISTRY/2112	+NAME+ADDRESS

Entity	Step	Virtual entity 1 records	Virtual entity 2 records	Match key
300005	1	VENDOR_REGISTRY/2121	VENDOR_REGISTRY/2122	+NAME+PHONE
300007	1	VENDOR_REGISTRY/2131	VENDOR_REGISTRY/2132	+NAME+ADDRESS
300009	1	VENDOR_REGISTRY/2141	POLICYHOLDERS/2142	+NAME+ADDRESS
300010	1	VENDOR_REGISTRY/2151	POLICYHOLDERS/2152	+NAME+PHONE
300011	1	VENDOR_REGISTRY/2161	VENDOR_REGISTRY/2162	+NAME+ADDRESS+EMAIL
400001	1	SIU_WATCHLIST/1006	POLICYHOLDERS/1005	+NAME+DRLIC
400004	1	SIU_WATCHLIST/1014	POLICYHOLDERS/1009	+PNAME+DOB+SSN
400004	2	SIU_WATCHLIST/1014; POLICYHOLDERS/1009	POLICYHOLDERS/1010	+NAME+DOB+ADDRESS
400004	3	SIU_WATCHLIST/1012	SIU_WATCHLIST/1014; POLICYHOLDERS/1009; POLICYHOLDERS/1010	+NAME+DOB+ADDRESS
400004	4	SIU_WATCHLIST/1012; SIU_WATCHLIST/1014; POLICYHOLDERS/1009; POLICYHOLDERS/1010	POLICYHOLDERS/1011	+NAME+DOB+ADDRESS
400006	1	SIU_WATCHLIST/1021	POLICYHOLDERS/1020	+ADDRESS+SSN+EMAIL
400007	1	SIU_WATCHLIST/1024	POLICYHOLDERS/1022	+NAME+DOB
400007	2	SIU_WATCHLIST/1024; POLICYHOLDERS/1022	POLICYHOLDERS/1023	+NAME+DOB
400010	1	SIU_WATCHLIST/1037	SIU_WATCHLIST/1038	+PNAME+DOB+ADDRESS+EMAIL
400010	2	POLICYHOLDERS/1035	POLICYHOLDERS/1036	+NAME+DOB+ADDRESS
400010	3	POLICYHOLDERS/1034	POLICYHOLDERS/1035; POLICYHOLDERS/1036	+NAME+DOB+ADDRESS
400010	4	SIU_WATCHLIST/1037; SIU_WATCHLIST/1038	POLICYHOLDERS/1034; POLICYHOLDERS/1035; POLICYHOLDERS/1036	+PNAME+DOB+EMAIL
500001	1	POLICYHOLDERS/1002	POLICYHOLDERS/1004	+NAME+DOB+ADDRESS
500001	2	POLICYHOLDERS/1002; POLICYHOLDERS/1004	POLICYHOLDERS/1003	+NAME+DOB+EMAIL
500001	3	POLICYHOLDERS/1001	POLICYHOLDERS/1002; POLICYHOLDERS/1003; POLICYHOLDERS/1004	+NAME+DOB+PHONE
500009	1	POLICYHOLDERS/1015	POLICYHOLDERS/1016	+NAME+DOB+ADDRESS
500009	2	POLICYHOLDERS/1015; POLICYHOLDERS/1016	POLICYHOLDERS/1017	+NAME+SSN
500009	3	POLICYHOLDERS/1015; POLICYHOLDERS/1016; POLICYHOLDERS/1017	POLICYHOLDERS/1018	+NAME+DOB+PHONE
600002	1	SIU_WATCHLIST/1041	SIU_WATCHLIST/1042	+NAME+ADDRESS
700010	1	POLICYHOLDERS/1030	POLICYHOLDERS/1031	+NAME+DOB+ADDRESS
700012	1	POLICYHOLDERS/1032	POLICYHOLDERS/1033	+NAME+DOB+PHONE
700019	1	POLICYHOLDERS/1043	POLICYHOLDERS/1045	+NAME+ADDRESS+PASSPORT
700020	1	POLICYHOLDERS/1044	POLICYHOLDERS/1046	+NAME+ADDRESS+PASSPORT

Entity	Step	Virtual entity 1 records	Virtual entity 2 records	Match key
700023	1	POLICYHOLDERS/1047	POLICYHOLDERS/1048	+NAME+EMAIL
700023	2	POLICYHOLDERS/1047; POLICYHOLDERS/1048	POLICYHOLDERS/1049	+NAME+EMAIL
700026	1	POLICYHOLDERS/1050	POLICYHOLDERS/1051	+NAME+DOB+PASSPORT
700026	2	POLICYHOLDERS/1050; POLICYHOLDERS/1051	POLICYHOLDERS/1052	+NAME+PASSPORT
700029	1	POLICYHOLDERS/1053	POLICYHOLDERS/1055	+NAME+ADDRESS+PASSPORT
700030	1	POLICYHOLDERS/1054	POLICYHOLDERS/1056	+PNAME+DOB+ADDRESS+DRLIC
700035	1	POLICYHOLDERS/1059	POLICYHOLDERS/1060	+NAME+PHONE
700037	1	POLICYHOLDERS/1061	POLICYHOLDERS/1062	+NAME+PHONE
700039	1	POLICYHOLDERS/1066	POLICYHOLDERS/1067	+NAME+SSN
700039	2	POLICYHOLDERS/1065	POLICYHOLDERS/1066; POLICYHOLDERS/1067	+NAME+DRLIC
700039	3	POLICYHOLDERS/1064	POLICYHOLDERS/1065; POLICYHOLDERS/1066; POLICYHOLDERS/1067	+NAME+PASSPORT
700039	4	POLICYHOLDERS/1063	POLICYHOLDERS/1064; POLICYHOLDERS/1065; POLICYHOLDERS/1066; POLICYHOLDERS/1067	+NAME+DOB
700039	5	POLICYHOLDERS/1063; POLICYHOLDERS/1064; POLICYHOLDERS/1065; POLICYHOLDERS/1066; POLICYHOLDERS/1067	POLICYHOLDERS/1068	+NAME+ADDRESS
700049	1	POLICYHOLDERS/1073	POLICYHOLDERS/1074	+NAME+DOB+ADDRESS
700051	1	POLICYHOLDERS/1075	POLICYHOLDERS/1076	+NAME+DOB+ADDRESS
700053	1	POLICYHOLDERS/1077	POLICYHOLDERS/1078	+NAME+DOB+ADDRESS
700055	1	POLICYHOLDERS/1079	POLICYHOLDERS/1080	+NAME+DOB
700057	1	POLICYHOLDERS/1081	POLICYHOLDERS/1082	+NAME+DOB+ADDRESS+PASSPORT
700059	1	POLICYHOLDERS/1083	POLICYHOLDERS/1084	+NAME+DOB+ADDRESS+PASSPORT
700061	1	POLICYHOLDERS/1085	POLICYHOLDERS/1086	+NAME+DOB+PHONE+EMAIL
700063	1	POLICYHOLDERS/1087	POLICYHOLDERS/1088	+NAME+DOB+PHONE+EMAIL-GENDER
700067	1	POLICYHOLDERS/1091	POLICYHOLDERS/1092	+NAME+DOB+PHONE+EMAIL
700069	1	POLICYHOLDERS/1093	POLICYHOLDERS/1094	+PNAME+DOB+DRLIC
700071	1	POLICYHOLDERS/1095	POLICYHOLDERS/1096	+NAME+DOB+DRLIC
700073	1	POLICYHOLDERS/1097	POLICYHOLDERS/1098	+NAME+ADDRESS+EMAIL
700075	1	POLICYHOLDERS/1099	POLICYHOLDERS/1100	+NAME+EMAIL
700077	1	POLICYHOLDERS/1101	POLICYHOLDERS/1102	+NAME+EMAIL

Entity	Step	Virtual entity 1 records	Virtual entity 2 records	Match key
700079	1	POLICYHOLDERS/1103	POLICYHOLDERS/1104	+NAME+DRLIC
700082	1	POLICYHOLDERS/2031	POLICYHOLDERS/2032	+NAME+ADDRESS
700095	1	POLICYHOLDERS/2192	POLICYHOLDERS/2193	+NAME+DOB+ADDRESS
700103	1	POLICYHOLDERS/2207	POLICYHOLDERS/2213	+NAME+ADDRESS

The report contains 72 steps for 52 entities, and each step has a returned match key. The complete JSON audit, including rule codes, is reports/query/how_merge_audit.json.

Review queue

The SDK scan covered 90 entities and returned 14 distinct cross-source POSSIBLY_SAME pairs. Two pairs have the ambiguity flag set; the other twelve do not. Every pair is a human-review candidate, not an asserted missed match or a fraud determination.

Entity pair	Sources in entity A	Sources in entity B	Match key	Ambiguous flag
200004 / 300013	VENDOR_REGISTRY, POLICYHOLDERS	VENDOR_REGISTRY	+NAME	0
200007 / 300015	VENDOR_REGISTRY, POLICYHOLDERS	VENDOR_REGISTRY	+NAME	0
200008 / 300014	VENDOR_REGISTRY, POLICYHOLDERS	VENDOR_REGISTRY	+NAME	0
200009 / 600006	VENDOR_REGISTRY	SIU_WATCHLIST	+NAME	0
200010 / 600007	VENDOR_REGISTRY	SIU_WATCHLIST	+NAME	0
400001 / 400003	SIU_WATCHLIST, POLICYHOLDERS	SIU_WATCHLIST	+NAME	0
400001 / 500001	SIU_WATCHLIST, POLICYHOLDERS	POLICYHOLDERS	+NAME+ADDRESS-DOB	0
400002 / 700019	SIU_WATCHLIST	POLICYHOLDERS	+NAME	0
400002 / 700020	SIU_WATCHLIST	POLICYHOLDERS	+NAME	0
400003 / 500001	SIU_WATCHLIST	POLICYHOLDERS	+NAME	0
400008 / 700007	SIU_WATCHLIST	POLICYHOLDERS	+NAME+DOB	0
400009 / 700009	SIU_WATCHLIST	POLICYHOLDERS	+NAME+ADDRESS-GENERATION	0
600002 / 700017	SIU_WATCHLIST	POLICYHOLDERS	+NAME+ADDRESS (Ambiguous)	1
600002 / 700018	SIU_WATCHLIST	POLICYHOLDERS	+NAME+ADDRESS (Ambiguous)	1

Names are omitted from this compact queue to keep identifiers and review keys prominent. The JSON report includes the entity names and the same source membership: reports/query/cross_source_review_queue.json.

Why and how: worked examples

Why (resolved example): POLICYHOLDERS/1011 and SIU_WATCHLIST/1014 are both in Entity 400004. The SDK returned RESOLVED with +NAME+DOB+ADDRESS. The supporting comparisons were name Ed Knight / Eddie Knight (97, CLOSE), DOB 3/1/70 / 3/1/70 (100, SAME), and address 1602 Brenville Pl San Francisco CA 94105 / 160 Brenville Pl San Francisco CA 94105 (95, CLOSE). These are the reported feature comparisons for this resolution, not an independent adjudication.

How (merge sequence): Entity 400004 was built in four steps: SIU_WATCHLIST/1014 joined POLICYHOLDERS/1009 on +PNAME+DOB+SSN; POLICYHOLDERS/1010 joined that

group on +NAME+DOB+ADDRESS; SIU_WATCHLIST/1012 joined the group on +NAME+DOB+ADDRESS; and POLICYHOLDERS/1011 joined on +NAME+DOB+ADDRESS. The SDK's final How state contains all five records in one virtual entity with no reevaluation pending. See reports/query/how_analysis.json for the step evidence.

Near-miss (the one that teaches more):

> Entities 400001 and 500001 remain a POSSIBLY_SAME review lead under +NAME+ADDRESS-DOB, not a resolved entity. The comparison shows Robbie Smith / Robert Smith at 97 (CLOSE) and 123 Main St, Las Vegas / 123 Main Street, Las Vegas NV 89132 at 99 (SAME), but DOB 03/31/1954 / 12/11/1978 at 59 (NO_CHANCE). This is concrete conflicting evidence; it does not establish whether the records are the same person. Details are in reports/query/quality_samples.json.

Relationship networks

The SDK found a two-hop path from Entity 200008 (Universal Exports Worldwide; VENDOR_REGISTRY and POLICYHOLDERS) through Entity 200007 (Universal Exports USA; VENDOR_REGISTRY and POLICYHOLDERS) to Entity 300001 (Jones Kaylee; VENDOR_REGISTRY):

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Entity 200008 -> Entity 200007 -> Entity 300001
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The first disclosed link is supported by reference key 2074 (Global Parent) between VENDOR_REGISTRY/2071 and VENDOR_REGISTRY/2074. The second is supported by reference key 2071 (Principal) between VENDOR_REGISTRY/2071 and VENDOR_REGISTRY/2101. This is a path across explicit source relationships, not a single record's statement that the endpoints are directly connected. The complete network and path response is reports/query/relationship_networks.json.

What was not found, and why

Measurement:

> Distinct-value overlap measured from the improved source files: POLICYHOLDERS x VENDOR_REGISTRY shared 1 phone value and 0 first-name, last-name, DOB, and email values. POLICYHOLDERS x SIU_WATCHLIST shared 6 last names, 5 first names, 3 DOBs, 2 SSNs, 1 email, and 1 phone value. VENDOR_REGISTRY x SIU_WATCHLIST shared 0 phone, first-name, last-name, email, and DOB values. These are counts of distinct values on the named fields, not counts of matching records or entities; see docs/data_source_evaluation.md.

The available evidence points to the data had little overlap to find, especially for VENDOR_REGISTRY x SIU_WATCHLIST and POLICYHOLDERS x VENDOR_REGISTRY; POLICYHOLDERS x SIU_WATCHLIST does have measured shared values. The cross-source entity review corroborated 8 of 15 sampled entities and left 7 for human follow-up. This evidence does not establish absolute pipeline accuracy or prove that every possible pair should resolve.

The match-key suppressor audit must be read by its original bucket: single-source per-record comparisons had 1 suppressor in 47 comparisons (GENDER); cross-source per-record comparisons had 0 in 24; relationship comparisons had 15 with suppressors in 70 comparisons (DOB 13, SSN 3, GENERATION 2, PASSPORT 1). Feature incidences can overlap, so those feature counts are not additive. A relationship suppressor records a declined entity merge; it is not the same as a per-record suppressor in records that were merged. The audit is documented in docs/results_validation.md and config/bootcamp_progress.json.

Additional constraints: the three source quality scores remain below the project's 70% threshold, with completeness of 33.2% for POLICYHOLDERS, 38.5% for VENDOR_REGISTRY, and 39.5% for SIU_WATCHLIST. Missing values were not inferred or imputed. Absolute accuracy remains unmeasured because there are no adjudicated truth labels.